

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 05)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

1. Write a Python program (named **Lab5_1.py**) to do the following:
 - Display a menu message (as shown in sample program output below)
 - Ask user to input a string or a character and store in a variable (in string type) named `uInc`
 - Display the input string or character on screen
 - Loop back to the beginning by `while` loop, until the character is "q" or "Q" → Quit the program.

User Input,
with <ENTER>

Sample Program Output

```
Please input a character or string, then <ENTER>; 'q'/'Q' for Quit: a
The character or string you typed is < a >
Please input a character or string, then <ENTER>; 'q'/'Q' for Quit: 3
The character or string you typed is < 3 >
Please input a character or string, then <ENTER>; 'q'/'Q' for Quit: abc
The character or string you typed is < abc >
Please input a character or string, then <ENTER>; 'q'/'Q' for Quit:
The character or string you typed is <  >
Please input a character or string, then <ENTER>; 'q'/'Q' for Quit: Q
The character or string you typed is < Q >
The program quits, as you want...
```

2. Write a guessing game program (named **Lab5_2.py**) to do the following: (Use `while` loop)
 - Display a menu message (as shown in sample program output below)
 - Ask user to input a number and guess your predefined “lucky” number (e.g. 88, within 1 to 100)
 - Store the input number and predefined lucky number in two different variable named `uInt` and `luckyNum`
 - Match the user input number and your predefined lucky number; loop back to the beginning to display the right message and guess the lucky number again
 - Quit the program until the right lucky number is entered

Sample Program Output

```
Listing divisible integers between 1 and 500
Enter the first Integer, then <ENTER>: 33
Enter the second Integer, then <ENTER>: 151
Integers (1 to 500) divisible by 33 or 151 are:
33 66 99 132 151 165 198 231 264 297 302 330 363 396 429 453 462
```

3. Write a guessing game program (named **Lab5_3.py**) to do the following: (Use `while` loop)
- Display a menu message (as shown in sample program output below)
 - Ask user to input a number and guess your predefined “lucky” number (e.g. 88, within 1 to 100)
 - Store the input number and predefined lucky number in two different variable named `uInt` and `luckyNum`
 - Match the user input number and your predefined lucky number; loop back to the beginning to display the right message and guess the lucky number again
 - Quit the program until the right lucky number is entered

Sample Program Output

```
Guess our lucky number <between 1 and 100>, then <ENTER>: 99
The number you typed is [ 99 ] - Too LARGE, guess the number again!
Guess our lucky number <between 1 and 100>, then <ENTER>: 50
The number you typed is [ 50 ] - Too SMALL, guess the number again!
Guess our lucky number <between 1 and 100>, then <ENTER>: 75
The number you typed is [ 75 ] - Too SMALL, guess the number again!
Guess our lucky number <between 1 and 100>, then <ENTER>: 88
Right! This is our LUCKY NUMBER [ 88 ]. Well Done... QUIT!
```

~ END ~